Code-Layout, Readability and Reusability

Date	05 July 2026
Team ID	XXXXXX
Project Name	Smart Lender - AI Powered Loan Eligibility Prediction
Maximum Marks	5 Marks

This document evaluates the quality of the Smart Lender codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended in future development.

Code Layout Checklist

S.No	Code Quality Parameter	Remarks	Followed (Yes/No/Partial)
1	Consistent Indentation	Uniform 4-space indentation used throughout app.py and notebooks	Yes
2	Proper File Structure	Separate datasets/, models/, notebooks/, templates/ and static/ folders	Yes
3	Meaningful Variable Names	Variables such as applicant_income, credit_history reflect their purpose	Yes
4	Function / Route Names	Flask routes home() and predict() are descriptively named	Yes
5	Code Comments	Inline comments explain input parsing, encoding and prediction logic	Yes
6	Modular Design	Preprocessing, training and evaluation are split across separate notebooks	Yes
7	No Redundant Code	Duplicate logic minimized; encoders/scaler reused via Joblib	Yes
8	Error Handling	predict() route wraps prediction logic in a try/except block	Yes

Reusable Components / Modules

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Prediction Route (predict())	Python / Flask	Web request handling	High
2	StandardScaler (scaler.pkl)	Scikit-learn	Feature scaling before prediction	High
3	Label Encoders (label_encoders.pkl)	Scikit-learn	Encoding categorical inputs	High
4	Trained KNN Model (loan_model.pkl)	Scikit-learn	Loan status prediction	High
5	HTML Templates (index.html, result.html)	HTML/Bootstrap/Jinja 2	Form input and result display	Medium
6	style.css	CSS3	Consistent glass-card visual theme across pages	Medium

Overall Code Quality Assessment

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation between data, model and web layers
Readability	5	Descriptive naming and consistent formatting
Reusability	5	Preprocessing artifacts are reused directly at inference time
Documentation / Comments	4	Core logic is commented; more docstrings could be added
Overall Score	5	Production-ready structure for an academic project